MIL-H-80045B
29 July 1977
SUPERSEDING
MI L-H-80045A
29 April 1970

MILITARY SPECIFICATION

HEATERS, INDUCTION, HIGH FREQUENCY, VACUUM TUBE TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers induction type heaters utilizing vacuum tube oscillators to produce power in the radio frequency (RF) range.

1.2 Classification. The induction heaters shall be of the following types and sizes as specified (see 6.2.1).

Type I - Bench Mounted

Size 1 - 1 KW Power Output	Size 3 - 3 KW Power Output
Size 2 - 2 (W Power Output	Size 5 - 5 KW Power Output

Type II - Floor Mounted

Size 1 - 1 {W Power Output	Size 30 - 30 KW Power Output
Size 2 - 2 (W Power Output	Size 40 - 40 KW Power Output
Size 3 - 3 {W Power Output	Size 50 - 50 KW Power Output
Size 5 - 5 <W Power Output	Size 60 - 60 KW Power Output
Size 7 - 7 {W Power Output	Size 75 .- 75 KW Power Output
Size 10 - 10 {W Power Output	Size 100 - 100 KW Power Output
Size 15 - 15 {W Power Output	Size 150- 150 KW Power Output
Size 20 - 20 KW Power Output	Size 250 - 250 KW Power Output
Size 25 - 25 {W Power Output	Size 300 - 300 KW Power Output

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3424

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein,

SPECIFICATIONS

MILITARY

MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publ i cations required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw Thread Standards for Federal Services

FEDERAL COMMUNICATIONS COMMISSION (FCC)

Rules and Regulations, Vol. II, Part 18, Industrial, Scientific, and Medical Equipment

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments	Occupational Safety and Health Administration Standards

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

Standard No. 54 Standard Test Code, and Recommended Practice for Induction and Dielectric Heating Equipment

(Copies of IEEE Standards may be obtained from the Institute of Electrical and Electronic Engineers, 345 East 47th St., New York, NY 10017.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete heater for first article inspection and approval (see 4.2 and 6.4).

* 3.2 Design. The radio frequency induction heaters shall be designed for conversion of electrical energy at commercial power line frequencies to radio frequency energy. Unless otherwise specified, the nominal output frequency for heaters up to 75 KW shall be within the range of 400 through 450 KHZ, and for heaters rated at 100 KW and above, the nominal frequency shall be 300 KHZ. When specified, the output frequency shall be as specified (see 6.2.1). Frequency conversion shall be accomplished through the use of vacuum tube oscillator circuits. The units shall be designed and constructed to meet the requirements specified herein using load arrangements similar to those described in Standard IEEE No. 54, with a load circuit efficiency of not less than 75 percent and an overall efficiency of 45 to 50 percent. The heaters shall be specifically designed for industrial induction heating applications. Within rated power capacity, the units shall be capable of performing continuous production service without exceeding the rated working capacity of any components contained therein. Except as otherwise specified herein, the heaters shall be manufactured to comply with IEEE Standard No. 54 as it applies to radio frequency vacuum tube type induction heaters. The heaters furnished under this specification shall be new and of the manufacturers current design. All parts subject to wear shall be easily accessible for adjustment, replacement, and repair.

3.2.1 Safety devices.

3.2.1.1 Operator protection. The design of the heaters shall incorporate all necessary safety devices for the protection of operating and maintenance personnel. Electrical interlock safety switches shall be provided for all cabinet access doors and panels, except for bolted panels, the opening of which would expose personnel to voltages in excess of 120 volts. The design of the work or load coil circuitry shall be such that operating personnel will not be exposed to dangerous electrical shock. One side of the work coil or the transformer secondary shall be grounded and suitable terminals shall be provided for electrically grounding the entire cabinet and chassis assembly. Warning labels shall be attached to the unit at appropriate and

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conspicuous places to warn personnel when there is access, by removing doors or panels, to voltages above 250 volts. Capicators larger than 0.1 microfarad in circuits with potentials in excess of 230 volts shall be provided with automatic means of rapidly discharging the stored charge when the power is removed, Grounding sticks or an automatic grounding system shall be provided. When grounding sticks are furnished, grounding points shall be identified. Bleeder resistors shall be incorporated in auxiliary power supplies using filter capacitors in the output circuit. All control circuits shall be operated from a 115 volt AC power source, Except as otherwise specified by the procuring activity, the heater shall comply with the standards promulgated under OSHA applicable to the heater itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

3.2.1.2 Equipment protection. Circuit design and protective devices shall be incorporated to protect the oscillator plate and grid circuits and the DC power supplies" from currents in excess of the rated capacity of components contained therein. Protection shall be provided to prevent damage to the equipment from power line under voltage including under voltage in any phase of a three phase system. A circuit breaker type main power disconnect shall be provided for line overload and short circuit protection. A time delay device shall be incorporated to insure an adequate preheating period before high voltage is applied to the plate of the oscillator tube or tubes, and to the plates of the rectifiers when mercury vapor vacuum tube type rectifiers are used. Surge suppressors shall be placed across each phase of the high voltage transformer to prevent damage resulting from excessive voltage surges. A thermostatic switch shall be provided to remove both filament and plate voltage to the oscillator circuit when the water temperature in the cooling system exceeds the safe operating limit. A differential water pressure or water flow switch shall be incorporated to prevent operation of the unit when the water flow is interrupted or is insufficient to insure proper cooling of the heater components: A water strainer shall be provided-for use in the cooling system to prevent obstruction of the water passages by foreign matter. The strainer shall be designed so that it may be easily removed for cleaning or replacement. The strainer shall be either installed in the heating unit or furnished complete with all necessary fittings for external installation. On air cooled units, an ambient air temperature thermal limit switch and an air flow switch shall be provided to remove the power from the filament and the plate of the oscillator circuit when either the air temperature or the air flow (CFM) inside the cabinet deviates from a predetermined normal level for operation at full rated power output. To the maximum extent possible, all electrical components shall be mounted in a location that will Provide protection from water leaks or condensation that might develop in-the water cooling system.

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3.2.2 Lubrication. All moving parts requiring lubricants shall be provided with means for applying adequate lubrication. All lubrication points shall be accessible.

3.2.3 Interchangeability. All parts shall remanufactured to such standards as will permit replacement or adjustment without modification of parts or the heater.

3.2.4 Material. All material used in the manufacture of the induction heaters shall be of the quality necessary to produce a heater to meet the requirements described herein.

3.3 Construction. The heaters shall be constructed so that when installed and connected to the specified power and coolant source, they can be used for any purpose for which they were designed. The heaters shall be constructed of parts that are new and sound in structure, Repaired parts shall not be used.

3.3.1 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where those operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

3.3.2 Fastening devices. All screws, pins, bolts, and similar parts shall be Installed with means for preventing loss of tightness. Such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.3 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.3.4 Painting. Unless otherwise specified (see 6.2.1), the heater shall be painted and finished in accordance with the manufacturer’s standard commercial practice.

3.3.5 Threads. Unless otherwise specified (see 6.2.1), all threaded parts sahll conform to Handbook H28.

3.4 Performance. The high frequency induction heaters covered by this specification shall be capable of heating ferrous and non-ferrous metals to any desired temperature within the limits of the frequency and power output specified. The power factor shall be not less than 86 percent under full load conditions. The heaters shall be capable of performing all types of high frequency induction heating such as annealing, ‘stress relieving,

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brazing, soldering, hardening, and melting. The rated RF power output of the heaters shall be expressed in kilowatts (KW) available at the work station under continuous duty operation, The rated output shall be not less than that shown for Types and Sizes in 1.2.

3.5 Components.

3.5.1 Controls. The heater controls shall consist of no less than a main power circuit breaker, plate power off-on contactor, variable grid excitation control, plate overload reset control, grid overload reset control (unless otherwise protected), and output power control. Except for the main power circuit breaker, all controls shall be located on the front of the heating unit and mounted so that they are easily accessible to the operator. All controls shall be identified by permanent marking in accordance with the manufacturer's standard commercial practice. When specified (see 6.2.1), a foot off-on control switch for alternate control of output power shall be provided. The foot switch shall be equipped with a standard length cable, suitable for the purpose. Connection of the foot switch to the heater shall be by a quick disconnect plug and receptacle. Power "off" devices shall be in accordance with IEEE Standard No, 54.

3.5.2 Output power control. Unless otherwise specified (see 6.2.1), stepless thyratron or silicon controlled rectifier (SCR) power control shall be utilized. Control of the output power to the load shall be continuously variable from zero to 100 percent of rated output, On three phase heaters, the control shall be manual and have the capability of accepting a standard DC input signal from a temperature controller of either O-5 or 4-9 milliamperes (Ma).

3.5.3 Timer. A timer shall be provided for automatically controlling the heating cycle. The timer shall be calibrated in increments of one second, from zero through a minimum of sixty seconds, with a repeat accuracy of plus or minus 0.25 percent. When specified (see 6.2.1), an electronic keying device for precision control of short heating cycles shall be provided. The heater shall be designed for both manual or automatic control of the heating cycle.

3.5.4 Voltage regulation. Filament voltages shall be maintained within Plus or minus 2 Percent of rated value for Plus or minus 10 percent line voltage variation. When specified (see 6.2.1), overall line-voltage or oscillator plate voltage regulation shall be provided. Regulation of plus or minus 2 percent of rated voltage shall be maintained with a line voltage variation of plus or minus 10 percent. If voltage regulation is not provided, filament and plate supply transformers shall be provided with taps to be used for adjusting the voltage to the filaments of the rectifier and oscillator tubes, and the plate voltage to the oscillator tube. The transformer taps shall be in accordance with IEEE Standard No. 54.

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3.5.5 Wiring and wiring devices. All wiring and wiring devices shall be in accordance with IEEE Standard No. 54.

3.5.6 Load matching. An adjustable tank inductance shall be provided to permit quick and convenient resonating over a wide range of load conditions. When specified (see 6.2.1),-a radio frequency output transformer for matching the output of the oscillator to the load shall be furnished.

3.5.7 Heating stations. The output terminals for the heater coils shall be mounted on the front of the heater and shall be designed for quick, convenient changing of coils. Unless otherwise specified (see 6.2.1), each heater shall be equipped for operation with one heating station.

3.5.8 Electrical indicating instruments. Individual meters shall be provided for indicating the values of the oscillator direct current plate voltage, plate current, and grid current. An hour meter to record the hours of filament operation for the oscillator and rectifier tubes shall be furnished, A radio frequency ammeter or an output power meter shall be included to assist in adjustment for maximum power output and efficiency. All direct current meters shall have a full scale accuracy of plus or minus 2 percent, and alternating current meters a full scale accuracy of plus or minus 3 percent. All meters except the hour meter shall be mounted and positioned on the front of the heater for easy reading by the operator. The hour meter shall be mounted in a convenient location either inside or outside the cabinet. The size of the meters shall be not less than 2-1/2 inches diameter for units up to 5 KW and not less than 4 inches diameter for units larger than 5 KW. All meter scales shall indicate directly in volts or kilovolts (KV), watts or KW, amperes, hours or milliamperes, as applicable, and shall be permanently marked and graduated in accordance with standard commercial practice. The function of all meters shall be identified by permanent marking in accordance with the manufacturer’s standard commercial practice.

3.5.9 Indicating lights. Pilot lights shall be installed on the front of the heater to monitor the following control functions and Protective devices: filament on, filament ready, heat on, and plate overload; cooling system water flow and cooling system water temperature, as applicable. The function of all indicating lights shall be identified by permanent marking in accordance with the manufacturer’s standard commercial practice.

3.5.10 Cooling system. Except for air cooled units, cooling of the heater shall be accomplished by a non-ferrous water cooling system and by filtered air circulation. The unit shall be designed for cooling with built-in water cooling system for connection to available city water supply.

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The system shall be designed so that any moisture appearing on the water cooled components as the result of condensation shall be disposed of in such a manner as not to be detrimental to proper functioning of the heater. All seals, gaskets, connections, tubing, and hoses shall be constructed of heavy duty material, designed and installed to prevent leakage. The air cooling system shall maintain positive pressure inside the cabinet or shall be a self-cooled totally enclosed system as specified in 3.5.11. The system shall provide sufficient air circulation for adequate cooling of all air cooled components when operating at full rated power. When specified (see 6.2.1), each heater shall be equipped with a self-contained closed water cooling system utilizing a water to water heat exchanger suitable for use with water supplied from a storage tank, plant, or city water system. The cooling system shall be adequate for cooling the unit when operating under "usual service conditions" as defined in IEEE Standard No. 54.

3.5.11 Cabinet and chassis. The cabinet shall be of aluminum or steel frame and panel construction. The gage of the metal and the physical strength of the cabinet and chassis, shall be adequate for support and protection of all components. The cabinet shall be constructed for minimum radiation of RF power. The cabinet shall be designed for operation with sufficient positive internal air pressure to assure that outside air is brought in only through a replaceable type air filter or shall be totally enclosed, self-cooled, not requiring ventilation.

3.5.12 Suppression of radio frequency radiation. Design and construction of the RF generating and radiating circuits shall provide the necessary harmonic suppression, shielding, and filtering to prevent radiation intensity in excess of that permitted by FCC Rules and Regulations, Volume II, Part 18.

3.5.13 Moisture and fungus-resistant treatment. When required (see 6.2.1), fungus and moisture resistant treatment shall be as specified.

3.6 Size and capacity. The Type I heaters shall be low power output heaters, and normally are designed for mounting and operating on a work bench. The Type II heaters shall be either low or high power heaters that are built in floor mounted cabinets. The size numbers designate the RF power output of the heaters in kilowatts.

3.7 Electrical input power and equipment requirements. The heaters shall be designed for operation from the alternating current (AC) power source specified (see 6.2.1). All three phase heaters shall be designed for three wire, three phase input. Performance shall be in accordance with requirements specified herein, with an input voltage variation of plus or minus 5 percent of rated input voltage. All electrical components including starters, relays, and switches shall have a rated continuous

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duty current carrying capacity equal to or greater than the maximum current flow to be encountered in the circuits involved. The voltage rating of all insulation shall equal or exceed the maximum voltage to which the components will be subjected. The heaters shall be completely wired through the controls and ready for connection to the applicable power source.

3.7.1 Motors. Motors shall be designed for continuous duty operation. Motor horsepower and speed shall be designed for maximum load operating conditions. Unless otherwise specified (see 6.2.1), motors shall be single speed and shall have a dripproof or better enclosure,

3.8 Equipment.

3.8.1 Standard equipment. Unless otherwise specified (see 6.2.1), each heater shall be furnished with all standard equipment normally furnished by the manufacturer, including one set of any 'special tools normally furnished with the heaters,

3.8.2 Optional equipment. When required (see 6.2.1), optional equipment shall be furnished as specified.

3.8.3 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.9 Electromagnetic interference control. When specified (see 6.2.1), the heaters shall have electromagnetic interference control in accordance with the applicable requirements-of MIL-STD-461.

3.10 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate shall be securely attached to each heater. The nameplate shall contain the information listed below. If the heater is a special model, the model designation shall include the model of the basic standard heater, and a suffix keyed to the manufacturer's permanent records,

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input characteristics and ratings
Rated power output - KW
Output frequency - KHz or MHz
Date of Manufacture
Contract or Order Number
National Stock Number or Plant Equipment Code
u s

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3.11 Technical data. Technical data shall be furnished as specified (see 6.3).

3.12 Workmanship. Workmanship of the heater and accessories shall be commensurate with the requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the first article when required in accordance with 3.1. Unless otherwise specified (see 6,2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the first article to pass the examination or any first article test shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each heater prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the tests in 4.6.1, and the inspection in 4.7. Failure of the heaters to pass the examination, test, or inspection shall be cause for rejection.

4.5 Examination. The heaters and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests. The test requirements indicated below shall supplement but not supersede or replace the manufacturer's standard equipment test procedures.

4.6.1 Operational test. With rated electrical input power and coolant applied, each heater shall be tested as specified below.

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4.6.1.1 Power output. The minimum rated continuous duty RF power output shall be tested to verify conformance to 3.4, This test shall be performed as described in IEEE Standard No. 54, Section III, “Test Code”.

4.6.1.2 Controls, instrumentation, and safety devices. All controls, meters, indicating lights, equipment protective and safety devices shall be functionally tested to verify correct operation.

4.6.1.3 Cooling system. The cooling system shall be tested to verify adequate cooling of the heaters when operating at maximum rated power. Water cooling systems shall be inspected for leaks.

4.6.2 Performance test. Performance tests shall be performed as described below.

4.6.2.1 Overall efficiency test. This test shall be performed to verify conformance to 3.2. The test shall be performed as described in IEEE Standard No. 54, Section III.

4.6.2.2 Load circuit efficiency test. This test shall be performed to verify conformance to 3.2. he test shall be performed as described in IEEE Standard No. 54, Section III.

4.6.2.3 Radiation test. Radiation tests shall be performed as necessary to verify that harmonic suppression, RF shielding, and filtering is adequate to assure performance within FCC regulations when the heaters are properly installed for use (see 6.7).

4.6,3 Electromagnetic interference control test. Heaters having electromagnetic interference control shall be tested in accordance with the applicable requirements of MIL-STD-461 to determine compliance with 3.9.

4.7 Inspection of preparation for delivery. Preservation, packing, and marking for shipment and storage shall be inspected to determine conformance to the requirements of Section 5.

5. PACKAGING

5,1 Preservation, packing and marking. Preservation, packing, and marking shall be in accordance with applicable requirements of MIL-E-17555. Degree of preservation and packing shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The radio frequency induction heaters are to be used for hardening, stress relieving, normalizing, melting, soldering, brazing, and other metal heat treating operations.

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6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.
b. Type and size required (see 1.2).
c. First article, when required (see 3.1),
d. Output frequency range, if different (see 3.2).
e. Specify exceptions and additional safety and health requirements (see 3.2.1.1 and 6.5).
f. Painting, if different (see 3.3.4).
g" Threads required, if different (see 3.3.5).
ho Foot switch, if required (see 3.5.1),
i. Type of output power control required, if different (see 3.5.2),
j. Electronic keying device, if required (see 3.5.3).
k. When line voltage or oscillator plate voltage regulation is required (see 3.5.4).
l. Radio frequency power output transformer, when required (see 3.5.6).
m. Number of heating stations required, if different (see 3.5.7).
n. Self-contained closed coolant system, if required (see 3.5.10).
00 Moisture and fungus resistant treatment, if required (see 3.5.13).
p* Specify AC input voltage rating required (see 3.7).
q. Motor requirements, indifferent (see 3.7.1).
r. Standard equipment, if different (see 3.8.1).
s, Optional equipment, when required (see 3.8.2).
t. Repair parts required (see 3.8.3).
u. Electromagnetic interference control, when required (see 3.9).
v. Nameplate, if different (see 3.10).
w, First article inspection, if different (see 4.3 and 6.7).
w, Quality conformance inspection, if different (see 4.4).
y. Degree of preservation and packing required (see 5.1).

* 6.3 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

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* 6.4 First article, When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. A first article comprises a preproduction item or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, test and approval of the first article.

* 6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

* 6,6 Mercury prohibited, When the use of mercury or its compounds in the heater and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

6.7 Radio frequency radiation test. The final testing of this equipment to determine conformance to FCC requirements relative to maximum permissible radiation, can be performed only after installation in the user facility. Therefore, it is suggested that contracts awarded for this equipment provide for the services of-a contractor's engineer to supervise installation-and to perform the tests and certification required by FCC Rules and Regulations, Volume II, Part 18.

6.8 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.9 Definitions. Unless otherwise indicated herein, definition of terms shall be in accordance with IEEE Standard No, 54.

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User Activities:

Navy -MC, SH, AS
Army - SM, MU

Preparing Activity:

DSA- IP

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